Media Complexity Around Geopolitical Events: An Event-Based NLP Analysis of New York Times Coverage

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Abstract

This project studies how linguistic complexity changes around major U.S.-involved geopolitical events using *New York Times* coverage. We build a pipeline that collects articles from the NYT Article Search API, cleans and filters text, constructs interpretable NLP features (readability and lexical rarity), and estimates event-based pre/post fixed-effects regressions. We support regression output with event-level and domain-level figures and a placebo test. Across outcomes, point estimates suggest modest post-event simplification, but results exhibit meaningful heterogeneity across events and limited precision due to the small analytic sample after quality filtering.

Contents

1 Introduction **3**

2 Theoretical Motivation **3**

3 Data Collection **3**

3.1 Event Selection 3

3.2 NYT API Pipeline 4

4 Text Cleaning and Preprocessing **4**

5 Natural Language Processing Features **5**

5.1 Readability Metrics 5

5.2 Lexical Rarity 6

5.3 Rare-Word Share 6

5.4 Measurement Scope and Computational Considerations 7

5.5 Why These Measures? 7

6 Empirical Model **7**

7 Descriptive Tables **8**

8 Results **10**

8.1 Readability 10

8.2 Lexical Rarity 10

8.3 Baseline Visual Evidence 10

9 Extended Empirical Results and Visual Analysis **12**

9.1 Event-Level Pre/Post Dynamics (Readability) 12

9.2 Event-Level Pre/Post Dynamics (Lexical Rarity) 13

9.3 Domain-Level Patterns 14

9.4 Rare Word Share 14

9.5 Readability vs. Lexical Rarity (manual scatter) 15

9.6 Synthesis of Empirical Findings 16

10 Placebo Tests **16**

11 Limitations **17**

12 Extensions **17**

13 Conclusion **17**

1 Introduction

Major geopolitical events such as military interventions and terrorist attacks are moments of heightened public attention. During these periods, mass media outlets play a central role in shaping how complex events are communicated to the public. While prior research has examined media framing, sentiment, and agenda-setting, comparatively less attention has been paid to how *linguistic complexity itself* evolves around such events.

This project studies how the linguistic complexity of news reporting changes *before and after* major U.S.-involved geopolitical events, using article-level data from the *New York Times*. We focus on three dimensions of textual complexity:

- Readability (Flesch–Kincaid grade level),
- Lexical rarity (average word rarity),
- Share of rare words within articles.

Using a custom data pipeline that combines web APIs along with natural language processing we estimate whether reporting becomes systematically simpler or more complex following major events. Our approach emphasizes transparency, reproducibility, and interpretability rather than complex machine learning models such as feedforward neural networks.

2 Theoretical Motivation

From a theoretical perspective, media complexity around major events is ambiguous. On one hand, heightened public interest may incentivize journalists to simplify language to reach broader audiences quickly. On the other hand, events involving international conflict, military strategy, and diplomacy may necessitate more technical language and specialized terminology.

Our working hypothesis is therefore not directional *a priori*. Nonetheless we ask:

> *Does the linguistic complexity of news coverage systematically change after major US foreign intervention conditional on event and outlet characteristics?*

3 Data Collection

3.1 Event Selection

We study eight major geopolitical events between 2009 and 2018:

- The Afghanistan troop surge (2009),
- The raid on Osama bin Laden (2011),
- NATO intervention in Libya (2011),

- The Syrian chemical weapons crisis (2013),
- ISIS airstrikes announced (2014),
- The Mosul offensive (2016),
- U.S. strikes on Syria (Shayrat) (2017),
- The Douma chemical attacks response (2018).

Each event is assigned a clearly defined date and a pre-specified window around the event. Articles are classified as `pre` or `post` based on whether their publication date falls before or after the event date (within the window).

3.2 NYT API Pipeline

Articles are collected using the New York Times Article Search API. For each event:

- A tailored Boolean query is constructed,
- Articles are retrieved within the event window,
- Robust retry logic with exponential backoff handles rate limits and network failures.

Listing 1: Event-driven NYT collection in the pipeline (excerpt).

```
# run_pipeline.py (excerpt)
events = load_events_nyt()
articles_raw = build_articles_index_nyt(events, per_event_cap=50)
articles_raw.to_parquet("data/interim/articles_raw_nyt.parquet", index=False)
```

Only articles with usable body text are retained. Duplicate URLs are removed to avoid over-counting syndicated content. The result is raw text with event identifiers and metadata necessary for later fixed-effects modeling.

4 Text Cleaning and Preprocessing

Raw article text is cleaned using a preprocessing pipeline:

1. Lowercasing,
2. Removal of punctuation and digits,
3. Whitespace normalization,
4. Tokenization and basic sentence segmentation.

Listing 2: Cleaning and minimum-word filtering (pipeline excerpt).

```
# run_pipeline.py (excerpt)
articles_text = clean_articles(articles_raw, min_words=60)
articles_text.to_parquet("data/interim/articles_text_nyt.parquet", index=False)
```

To ensure reliable readability and lexical statistics, we set a minimum of 60 words. After cleaning, the final analytic sample contains 58 articles from 350 pulled samples.

5 Natural Language Processing Features

This project utilizes *linguistic complexity* using interpretable, article-level text features derived from readability theory and frequency-based lexical analysis. Rather than relying on complex and unintuitive models, we construct low-dimensional measures that map directly to cognitive load and textual accessibility. This choice reflects the empirical goal of estimating interpretable treatment effects rather than maximizing predictive accuracy.

5.1 Readability Metrics

Our primary readability outcome is the *Flesch–Kincaid grade level* (FK), a widely used metric designed to approximate the U.S. school grade required to comprehend a text. For article i, the FK score is computed as:

$$\mathrm{FK}_i = 0.39 \cdot \overline{\mathrm{ASL}}_i + 11.8 \cdot \overline{\mathrm{ASW}}_i - 15.59,$$

where:

- $\overline{\mathrm{ASL}}_i$ denotes the average sentence length (words per sentence),
- $\overline{\mathrm{ASW}}_i$ denotes the average number of syllables per word.

Sentence segmentation and word tokenization are performed after text cleaning (lowercasing, punctuation removal, and whitespace normalization). Syllable counts are computed using a rule-based approach with standard implementations in applied readability research.

Listing 3: Readability feature construction (module call used in build_features).

```
# src/features.py (conceptual usage inside build_features)
from src.metrics import readability_scores

scores = readability_scores(text_clean)
fk_grade = scores["fk_grade"]
```

The FK grade is well-suited to this setting for three reasons. First, it is easily interpretable in units familiar to non-technical audiences. Second, it directly captures two core sources of processing difficulty longer sentences and longer words. Third, it varies meaningfully even within professional news writing, allowing us to detect relatively subtle shifts in accessibility over short time horizons.

5.2 Lexical Rarity

Readability metrics alone cannot distinguish between complexity arising from sentence structure and complexity arising from vocabulary choice. To capture the latter, we construct measures of *lexical rarity* based on word frequency.

Let w_{it} denote token t in article i, and let $f(w)$ denote the reference frequency of word w in a large text. We define the rarity score of a token as a monotonic transformation of frequency:

$$r(w) = -\log f(w),$$

so that lower-frequency words receive higher rarity scores. Article-level *average lexical rarity* is then defined as:

$$\text{avg_rarity}_i = \frac{1}{N_i} \sum_{t=1}^{N_i} r(w_{it}),$$

where N_i is the number of tokens in article i after punctuation removal.

Listing 4: Lexical rarity features (module call used in build_features).

```
# src/features.py (conceptual usage inside build_features)
from src.lingual import rarity_features

rar = rarity_features(text_clean)
avg_rarity = rar["avg_rarity"]
rare_share = rar["rare_share"]
```

This measure captures the extent to which an article relies on uncommon or specialized vocabulary, independent of sentence length or grammatical structure. In the context of news event reporting, higher average rarity may reflect the use of technical military terms or institutional jargon.

5.3 Rare-Word Share

To complement the continuous rarity measure, we also construct a threshold-based indicator of vocabulary concentration. Specifically, we define *rare word share* as:

$$\text{rare_share}_i = \frac{1}{N_i} \sum_{t=1}^{N_i} \mathbb{1}\{r(w_{it}) > \tau\},$$

where τ is a fixed rarity threshold corresponding to the upper tail of the rarity distribution.

Whereas average rarity reflects shifts in mean word choice, rare-word share captures whether an article contains a disproportionate mass of highly uncommon terms. An article can become more complex either by modestly increasing overall vocabulary sophistication or by introducing a small number of highly specialized terms.

5.4 Measurement Scope and Computational Considerations

Lexical rarity calculations are computationally more intensive than readability metrics. To ensure consistency rarity features are computed on a capped subset of articles when necessary. Sampling is performed uniformly across events to avoid introducing systematic bias in pre/post comparisons.

All text-derived features are computed at the article level and treated as continuous outcomes in regression analysis.

Listing 5: Feature construction and persistence (pipeline excerpt).

```
# run_pipeline.py (excerpt)
feats = build_features(
    articles_text,
    use_lingua=True,
    lingua_sample_cap=1500,
)
feats.to_parquet("data/features_nyt.parquet", index=False)
```

5.5 Why These Measures?

The choice of readability and rarity metrics is motivated by substantive and methodological considerations:

- **Interpretability:** Each measure corresponds to a clear linguistic mechanism: sentence length, word length, and word frequency.
- **Low dimensionality:** Numerical outcomes facilitate event-level fixed-effects estimation and visualization.
- **Causal compatibility:** These features can be meaningfully interpreted as outcomes in a pre/post event-study design.

In contrast to neural net/classification models, these measures allow us to directly quantify how accessible or specialized reporting becomes following major US foreign intervention events. This alignment between linguistic theory and econometric design is central to the empirical impact of the project.

6 Empirical Model

We estimate the following event-based pre/post fixed-effects model:

$$y_{ied} = \beta \cdot \text{PostEvent}_i + \alpha_e + \gamma_d + \varepsilon_{ied}$$

where:

- y_{ied} is a linguistic outcome for article i,
- PostEvent_i indicates post-event publication,
- α_e are event fixed effects,
- γ_d are domain/outlet fixed effects.

Standard errors are heteroskedasticity-robust (HC1). This specification isolates changes within events while controlling for event differences and domain differences.

Listing 6: Regression loop used in the pipeline (excerpt).

```
# run_pipeline.py (excerpt)
for outcome in ["fk_grade", "avg_rarity", "rare_share"]:
    model = run_event_prepost_regression(feats, outcome=outcome, use_domain_fe=True)
    print(model.summary())
```

7 Descriptive Tables

We first summarize the sample size and pre/post composition, then report mean outcomes by event and by domain.

Listing 7: Descriptive tables exported from features (post-pipeline script excerpt).

```
# post_pipeline_outputs.py (excerpt)
make_descriptive_tables(feats)  # writes tables/*.tex and tables/*.csv
```

Table 1: Counts summary for the analytic sample.

N_total	N_pre	N_post	N_events	N_domains
58	8	50	6	1

Table 2: Pre/Post means by event.

outcome	event_id	post_event	mean	n	std
fk_grade	AFG2009	0	13.5717	4	2.8603
fk_grade	AFG2009	1	10.8387	16	1.6735
fk_grade	IRQ2014	0	5.3165	1	–
fk_grade	IRQ2014	1	11.3724	4	2.0362
fk_grade	IRQ2016	1	16.6573	1	–
fk_grade	LBY2011	0	12.7938	3	0.4366
fk_grade	LBY2011	1	11.6700	6	1.0127
fk_grade	PAK2011	1	11.0398	17	1.6140
fk_grade	SYR2013	1	12.0657	6	2.8587
avg_rarity	AFG2009	0	6.9127	4	0.1660
avg_rarity	AFG2009	1	6.7720	16	0.1637
avg_rarity	IRQ2014	0	6.8461	1	–
avg_rarity	IRQ2014	1	6.9319	4	0.1427
avg_rarity	IRQ2016	1	8.4242	1	–
avg_rarity	LBY2011	0	6.9151	3	0.1642
avg_rarity	LBY2011	1	6.8127	6	0.2251
avg_rarity	PAK2011	1	6.8101	17	0.3076
avg_rarity	SYR2013	1	6.8842	6	0.1567
rare_share	AFG2009	0	0.1175	4	0.0119
rare_share	AFG2009	1	0.1106	16	0.0095
rare_share	IRQ2014	0	0.1018	1	–
rare_share	IRQ2014	1	0.1109	4	0.0097
rare_share	IRQ2016	1	0.1102	1	–
rare_share	LBY2011	0	0.1128	3	0.0053
rare_share	LBY2011	1	0.1094	6	0.0047
rare_share	PAK2011	1	0.1096	17	0.0087
rare_share	SYR2013	1	0.1099	6	0.0073
dale_chall	AFG2009	0	10.9941	4	0.6867
dale_chall	AFG2009	1	10.4528	16	0.6312
dale_chall	IRQ2014	0	7.6949	1	–
dale_chall	IRQ2014	1	10.5827	4	0.5319
dale_chall	IRQ2016	1	16.3215	1	–
dale_chall	LBY2011	0	11.0152	3	0.0880
dale_chall	LBY2011	1	10.7315	6	0.4169
dale_chall	PAK2011	1	10.1752	17	0.7544
dale_chall	SYR2013	1	10.9136	6	0.8433
avg_sentence_len	AFG2009	0	22.4829	4	3.9618
avg_sentence_len	AFG2009	1	16.9621	16	3.9661
avg_sentence_len	IRQ2014	0	11.4162	1	–
avg_sentence_len	IRQ2014	1	16.0210	4	4.6116
avg_sentence_len	IRQ2016	1	29.2500	1	–

Table 3: Pre/Post means by domain.

outcome	domain_simple	post_event	mean	n	std
$\text{fk}_g rade$	nytimes	0	12.2481	8	3.3990
$\text{fk}_g rade$	nytimes	1	11.3131	50	1.9152
$\text{avg}_r arity$	nytimes	0	6.9053	8	0.1417
$\text{avg}_r arity$	nytimes	1	6.8491	50	0.3194
$\text{rare}_s hare$	nytimes	0	0.1138	8	0.0099
$\text{rare}_s hare$	nytimes	1	0.1100	50	0.0081
$\text{dale}_c hall$	nytimes	0	10.5896	8	1.2540
$\text{dale}_c hall$	nytimes	1	10.5749	50	1.0802
$\text{avg}_s entence_l en$	nytimes	0	20.5377	8	4.7138
$\text{avg}_s entence_l en$	nytimes	1	17.8854	50	4.4923
ttr	nytimes	0	0.4565	8	0.1112
ttr	nytimes	1	0.4855	50	0.0965

8 Results

8.1 Readability

Post-event articles exhibit a negative but statistically insignificant change in Flesch–Kincaid grade:

$$\hat{\beta} = -1.02 \quad (p = 0.39)$$

This suggests a tendency toward simplification following events, though the estimate lacks statistical precision in the current sample.

8.2 Lexical Rarity

Average lexical rarity also declines slightly post-event:

$$\hat{\beta} = -0.098 \quad (p = 0.14)$$

Rare word share shows no meaningful change post-event in the pooled regression, consistent with only subtle shifts in word choice rather than a dramatic increase in specialized jargon.

8.3 Baseline Visual Evidence

Figure 1 shows pre/post FK grade by event from the pipeline output, while Figure 2 illustrates FK grade versus lexical rarity.

Listing 8: Baseline figures generated in the pipeline (excerpt).

```
# run_pipeline.py (excerpt)
plots.plot_outcome_pre_post_by_event(feats, outcome="fk_grade", figures_dir=FIGURES_DIR)
plots.plot_fk_vs_rarity_scatter(feats, figures_dir=FIGURES_DIR)
```

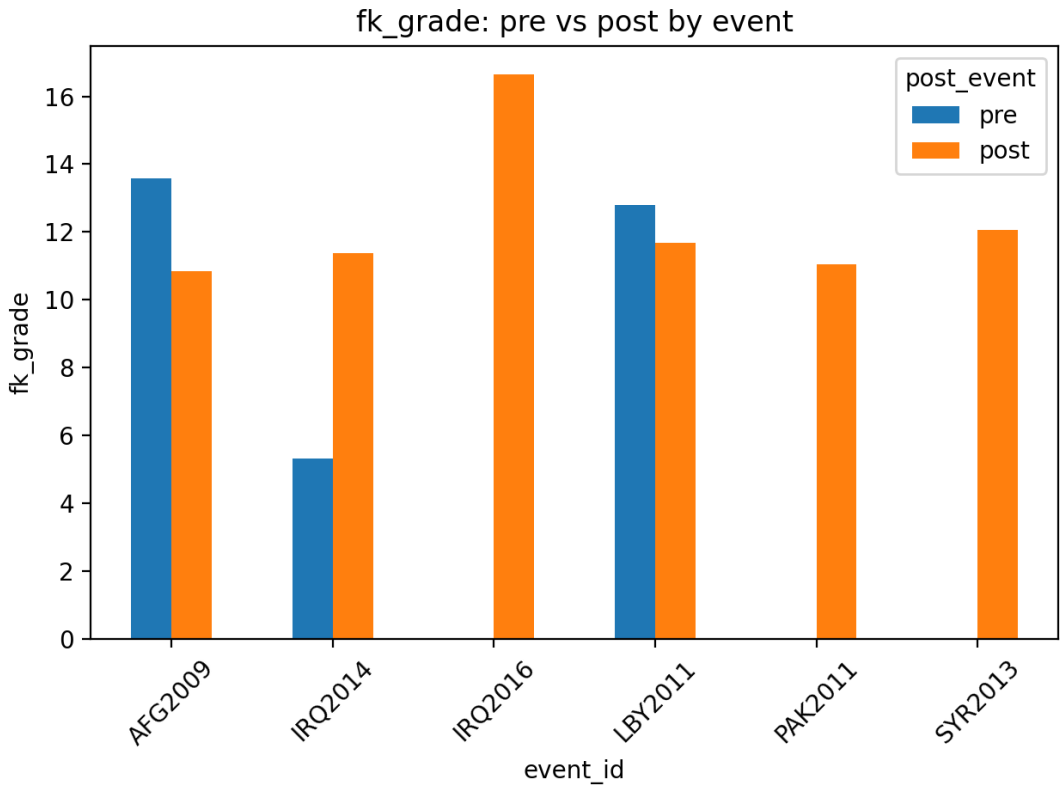

Figure 1: Pre/Post Readability by Event (pipeline figure)

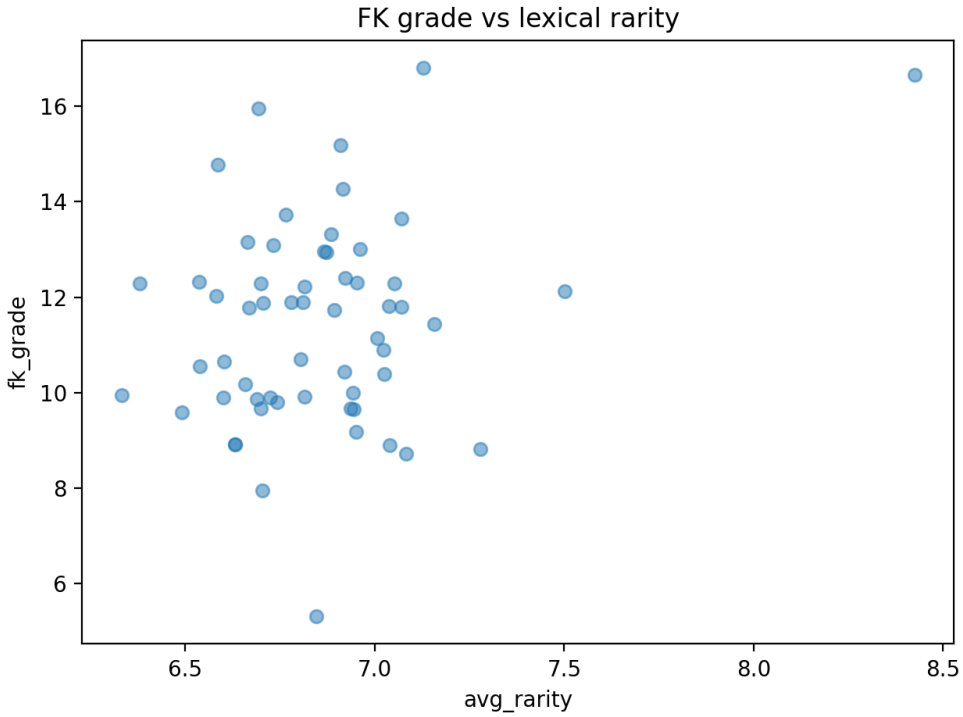

Figure 2: Readability vs. Lexical Rarity (pipeline scatter)

9 Extended Empirical Results and Visual Analysis

This section expands on the regression results by incorporating event-level and domain-level comparisons. These figures provide intuition for the regression estimates and reveal heterogeneity that can be masked in pooled coefficients.

9.1 Event-Level Pre/Post Dynamics (Readability)

Figure 3 plots mean Flesch–Kincaid grade levels before and after each event. Several patterns are apparent:

- Some events show post-event simplification (declines in FK grade), consistent with "explainer" coverage immediately after key developments.
- Other events move in the opposite direction, possible reflecting operational details or strategic reporting that becomes more prominent once an intervention is underway.
- The dispersion across events highlights why the pooled post-event coefficient can be small and statistically imprecise: post-event changes are not uniform.

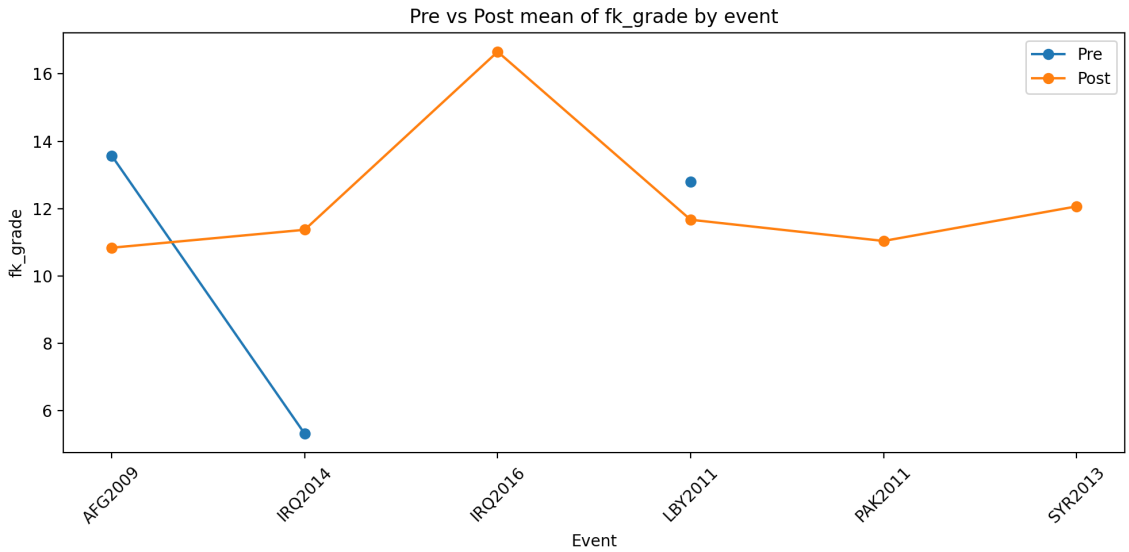

Figure 3: Pre/Post Mean Flesch–Kincaid Grade by Event

9.2 Event-Level Pre/Post Dynamics (Lexical Rarity)

Figure 4 shows analogous comparisons for average lexical rarity. Relative to FK grade, rarity movements are usually smaller, suggesting that shifts in readability can occur through sentence structure and organization even when average vocabulary rarity does not dramatically change.

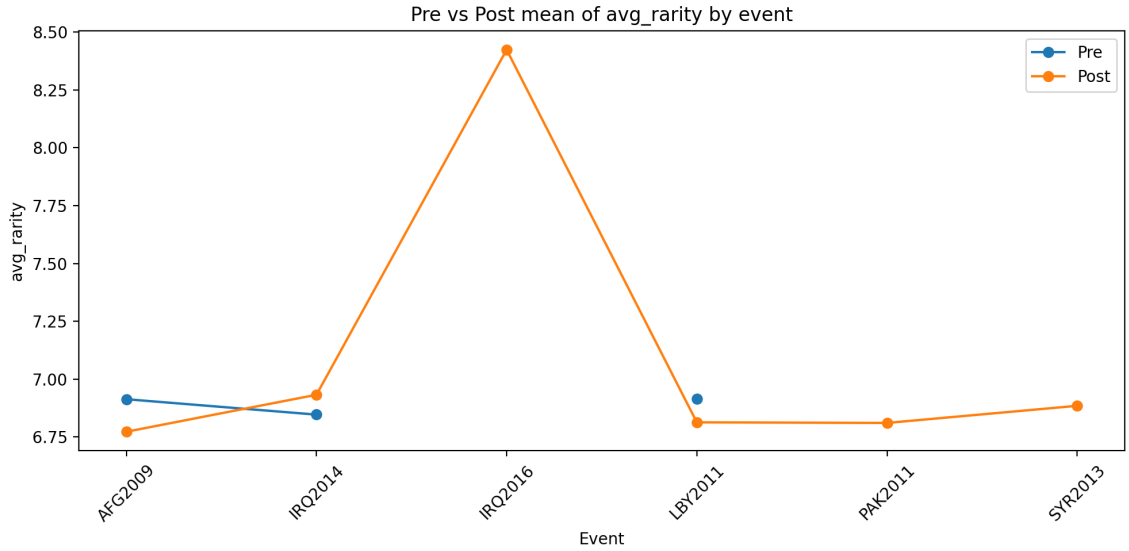

Figure 4: Pre/Post Mean Lexical Rarity by Event

9.3 Domain-Level Patterns

Figures 5 and 6 aggregate outcomes by domain. Because the sample is drawn from `nytimes.com`, domain comparisons primarily reflect internal publishing sections/feeds as captured in metadata. Both figures show modest post-event declines, consistent with the negative point estimates in the regressions.

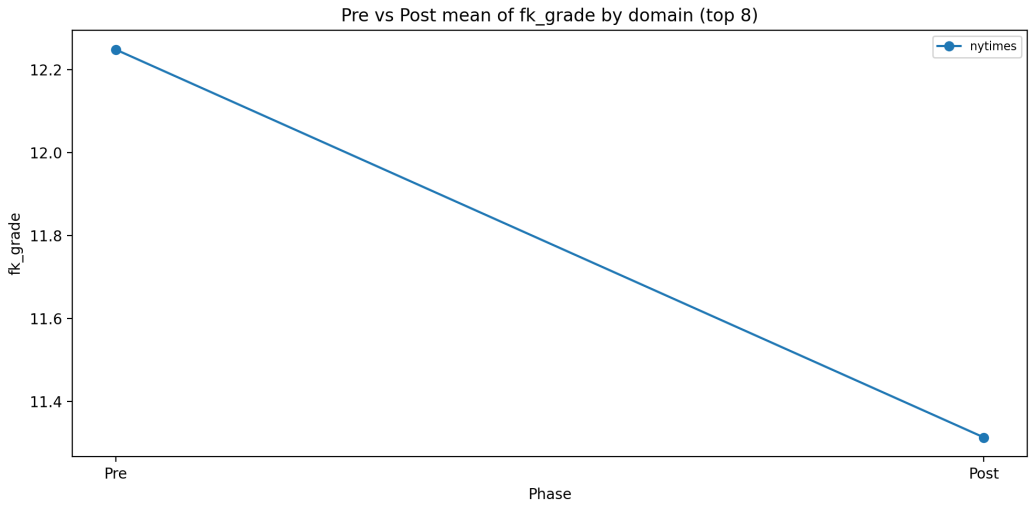

Figure 5: Pre/Post Mean Flesch–Kincaid Grade by Domain (Top 8)

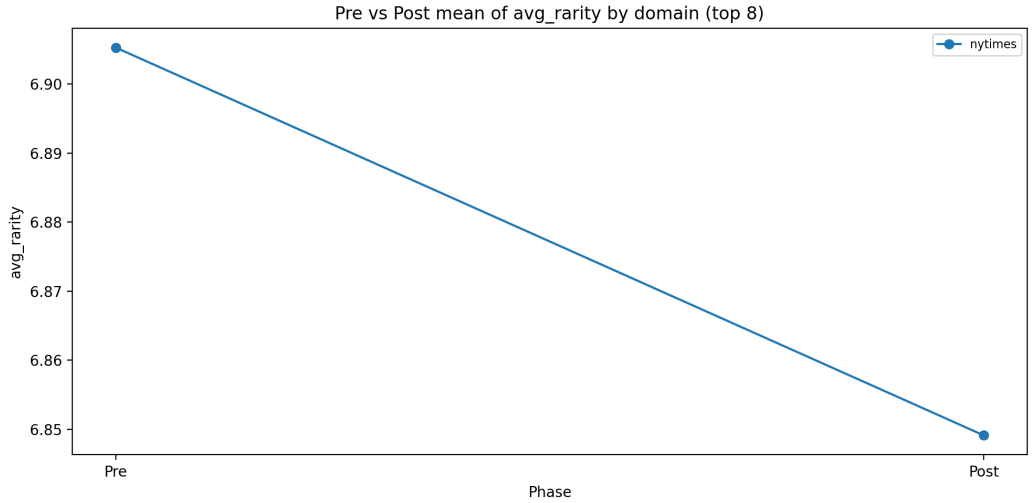

Figure 6: Pre/Post Mean Lexical Rarity by Domain (Top 8)

9.4 Rare Word Share

Rare word share exhibits minimal systematic change. Figures 7 and 8 show that while individual events display fluctuations, there is no consistent post-event increase in the proportion of rare

tokens. This suggests that post-event language shifts are driven more by incremental changes in typical word choice than by large-scale adoption of specialized terminology.

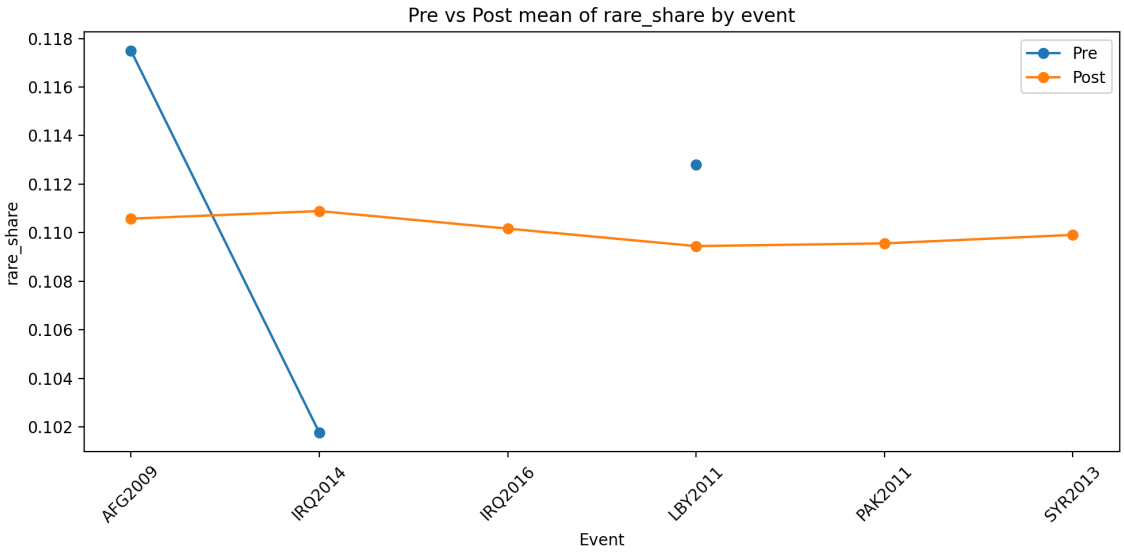

Figure 7: Pre/Post Mean Rare Word Share by Event

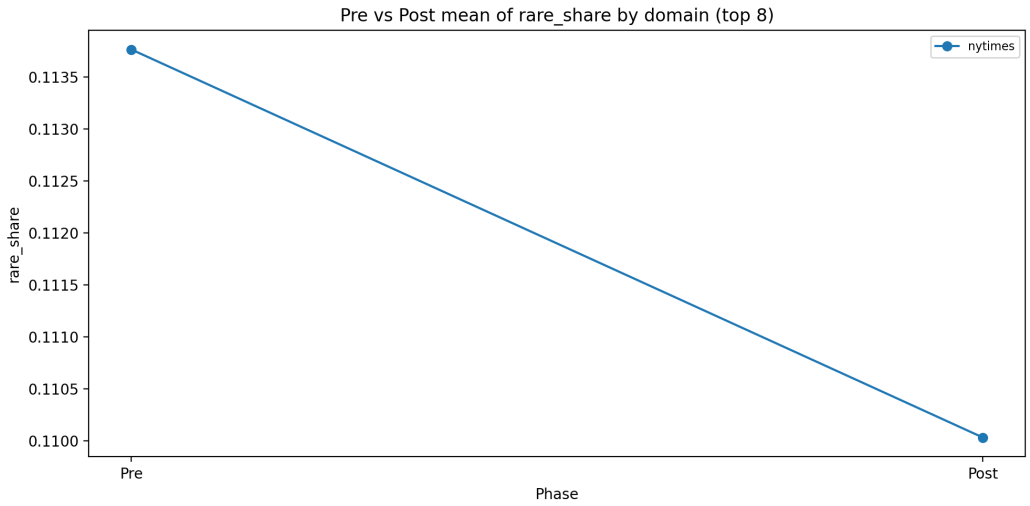

Figure 8: Pre/Post Mean Rare Word Share by Domain (Top 8)

9.5 Readability vs. Lexical Rarity (manual scatter)

Figure 9 plots article-level FK grade against average lexical rarity. The association is positive but noisy: articles with more uncommon vocabulary tend to be harder to read, but lexical rarity alone does not fully explain readability. This supports treating readability and rarity as complementary dimensions of complexity.

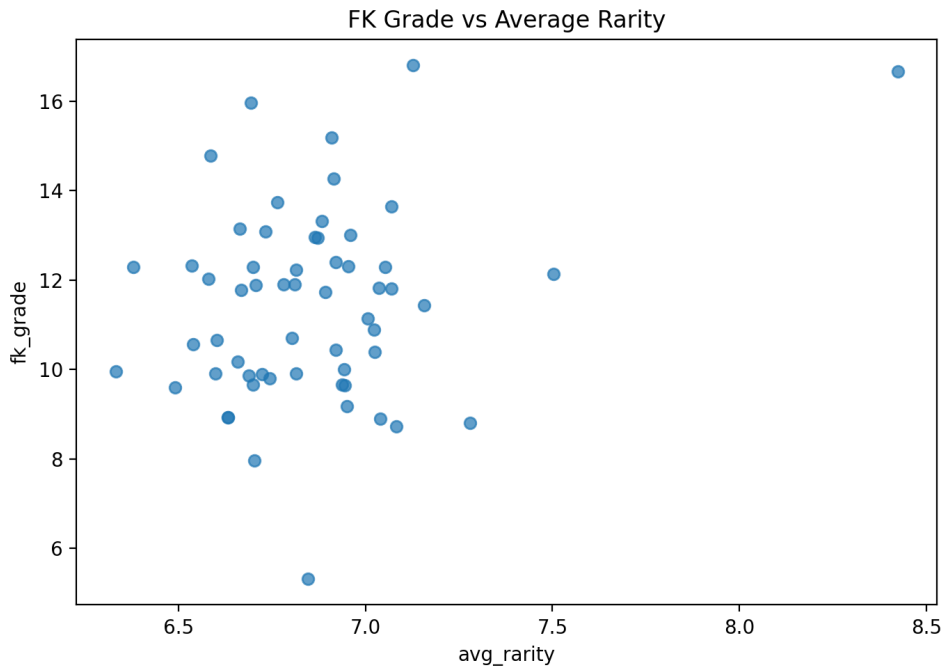

Figure 9: Flesch–Kincaid Grade vs. Average Lexical Rarity (manual scatter)

9.6 Synthesis of Empirical Findings

Taken together, the regression results and visual evidence suggest:

1. Linguistic complexity often declines after major US intervention events, consistent with simplified explanatory reporting and broad-audience writing immediately after shocks.

2. This pattern is not universal: some events move toward greater complexity post-event, consistent with detailed operational coverage, institutional jargon, and evolving tactical narratives.

3. Lexical rarity shifts are modest; when they occur, they appear to reflect subtle changes in average word choice rather than large increases in "rare-word density."

Given the small analytic sample after quality filtering, statistical insignificance should be interpreted as limited power rather than evidence of no relationship.

10 Placebo Tests

To validate the identification strategy, we conduct placebo tests by randomly assigning pseudo-event timing (within event windows) and recomputing the post indicator. The placebo coefficient distribution is centered near zero and is wide relative to the true estimate, which is consistent with the interpretation that observed effects are not purely mechanical artifacts of the design.

Listing 9: Placebo tests executed in the pipeline (excerpt).

```
# run_pipeline.py (excerpt)
placebo = run_placebo_tests(feats, outcome="fk_grade", n_placebo=200)
print(placebo.describe())
```

11 Limitations

Several limitations merit discussion:

- **Small sample size after filtering:** requiring usable text and a 60-word minimum reduces noise but limits power.
- **Text field limitations:** API-returned text may represent abstracts/snippets rather than full articles for some records.
- **Selection into publication:** the mix of article types post-event may differ from pre-event (breaking news vs. analysis).
- **Limited heterogeneity controls:** we do not explicitly control for topic shifts within event windows (battlefield updates vs. domestic politics).

These limitations motivate cautious causal interpretation and emphasize the value of this project as a pipeline and measurement contribution.

12 Extensions

Future work could:

- Expand to additional outlets and compare cross-outlet style changes,
- Incorporate framing measures or topic controls,
- Use embedding-based representations to study semantic abstraction,
- Link language shifts to engagement proxies when available.

13 Conclusion

This project demonstrates how modern NLP tools can be integrated with classical econometric techniques to study media behavior around major geopolitical events. While estimated effects are modest and not statistically strong in the current sample, the framework is flexible, interpretable, and easily extensible.

Overall, the evidence suggests a tendency toward slightly simpler language after major events, with meaningful heterogeneity across event types and only modest shifts in lexical rarity.

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